Boss Challenge — Paper 14 (Class 7)

Nutrition in Plants | Acids, Bases & Salts | Simple Equations | Rational Numbers

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

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1. The process by which green plants make their own food using sunlight is called:

(A) photosynthesis
(B) respiration
(C) transpiration
(D) digestion

2. The solution of the equation $x + 7 = 12$ is:

(A) $x = 84$
(B) $x = 5$
(C) $x = 7$
(D) $x = 19$

3. A substance that tastes sour and turns blue litmus red is best described as:

(A) a salt
(B) an acid
(C) an indicator
(D) a base

4. Which of the following is a rational number?

(A) only positive whole numbers
(B) a number that cannot be written as p/q
(C) -3/4
(D) any number with a square root sign

5. The green pigment in leaves that captures sunlight for photosynthesis is:

(A) haemoglobin
(B) starch
(C) chlorophyll
(D) cytoplasm

6. Solving $3x = 21$ gives the value of x as:

(A) $x = 63$
(B) $x = 18$
(C) $x = 7$
(D) $x = 24$

7. Bases generally taste bitter and feel:

(A) soapy
(B) gritty
(C) sour
(D) sticky

8. The standard (simplest) form of the rational number 12/18 is:

(A) 2/3
(B) 4/6
(C) 6/9
(D) 12/18

9. Tiny pores on the underside of a leaf through which gases enter and leave are called:

(A) roots
(B) chloroplasts
(C) stomata
(D) veins

10. The value of y in the equation y - 4 = 9 is:

(A) y = 9
(B) y = 13
(C) y = 36
(D) y = 5

11. A natural indicator extracted from lichen that is used to test acids and bases is:

(A) china rose
(B) litmus
(C) turmeric
(D) phenolphthalein

12. The sum 1/4 + 2/4 equals:

(A) 2/4
(B) 1/2
(C) 3/8
(D) 3/4

13. The raw materials a green plant needs to carry out photosynthesis are:

(A) starch and sunlight
(B) oxygen and glucose
(C) nitrogen and water
(D) carbon dioxide and water

14. Solving the equation x/5 = 4 gives:

(A) x = 9
(B) x = 0.8
(C) x = 1
(D) x = 20

15. Turmeric paper, a natural indicator, turns from yellow to which colour in a base?

(A) red
(B) blue
(C) green
(D) colourless

16. On the number line, the rational number -1/2 lies:

(A) halfway between 0 and -1
(B) to the right of 0
(C) exactly at 1
(D) halfway between 0 and 1

17. Plants like Cuscuta (Amarbel) that take food from another living plant are called:

(A) saprotrophs
(B) autotrophs
(C) insectivores
(D) parasites

18. Which equation correctly represents 'a number increased by 6 gives 15'?

(A) $6n = 15$
(B) $n - 6 = 15$
(C) $n + 6 = 15$
(D) $n + 15 = 6$

19. The reaction between an acid and a base that produces a salt and water is called:

(A) neutralisation
(B) condensation
(C) respiration
(D) evaporation

20. Which rational number is greater: 3/5 or 2/5?

(A) 3/5
(B) cannot be compared
(C) 2/5
(D) they are equal

21. A pitcher plant traps and digests insects mainly to obtain:

(A) sunlight
(B) glucose
(C) nitrogen
(D) water

22. The solution of $2x + 3 = 11$ is:

(A) $x = 2.5$
(B) $x = 7$
(C) $x = 8$
(D) $x = 4$

23. When an ant bites, it injects formic acid; the best home remedy is to rub on it:

(A) vinegar
(B) plain sugar
(C) a mild base like baking soda
(D) lemon juice

24. The additive inverse (opposite) of the rational number 5/7 is:

(A) 7/5
(B) 5/7
(C) 0
(D) -5/7

25. The mode of nutrition in which an organism feeds on dead and decaying matter is:

(A) parasitic nutrition
(B) autotrophic nutrition
(C) holozoic nutrition
(D) saprotrophic nutrition

26. If 5x = 35, then the value of x + 2 is:

(A) 37
(B) 12
(C) 7
(D) 9

27. Curd and lemon are sour because they contain, respectively:

(A) citric acid and lactic acid
(B) acetic acid and tartaric acid
(C) hydrochloric acid and nitric acid
(D) lactic acid and citric acid

28. The product 2/3 x 3/4 in simplest form is:

(A) 6/12
(B) 5/7
(C) 6/7
(D) 1/2

29. Rhizobium bacteria living in the root nodules of leguminous plants help by:

(A) absorbing extra sunlight for the plant
(B) storing water in the roots
(C) fixing nitrogen from the air into the soil
(D) digesting insects for the plant

30. The number which when multiplied by 4 and then 5 is added gives 25 is:

(A) 20
(B) 10
(C) 5
(D) 6

31. Which of these household substances is basic in nature?

(A) curd
(B) soap solution
(C) lemon juice
(D) vinegar

32. Between which two integers does the rational number 7/2 lie?

(A) 4 and 5
(B) 7 and 8
(C) 3 and 4
(D) 2 and 3

33. The test used to show that a leaf has made starch during photosynthesis uses:

(A) copper sulphate
(B) lime water
(C) blue litmus
(D) iodine solution

34. Solving $7 = x - 2$ gives:

(A) $x = 14$
(B) $x = 5$
(C) $x = 9$
(D) $x = 7$

35. When a base turns red litmus blue, we can say the solution is:

(A) neutral
(B) acidic
(C) basic
(D) salty

36. The rational number 0/9 is equal to:

(A) 9
(B) 1
(C) undefined
(D) 0

37. The food made during photosynthesis is mainly stored in plants in the form of:

(A) starch
(B) fat
(C) glucose
(D) protein

38. If one-third of a number is 8, the number is:

(A) 16
(B) 11
(C) 2.67
(D) 24

39. Excess use of fertilisers can make soil too acidic; to treat it, farmers add:

(A) vinegar
(B) more fertiliser
(C) common salt
(D) quicklime or slaked lime (a base)

40. Which sign makes the statement true: -2/3 ___ -1/3 ?

(A) cannot be decided
(B) > (greater than)
(C) = (equal)
(D) < (less than)

41. Photosynthesis can take place in the green stem of a plant because the stem contains:

(A) starch
(B) chlorophyll
(C) roots
(D) water

42. The equation for 'Reena is 3 years older than Sita, and Reena is 12' is:

(A) s - 3 = 12
(B) s + 12 = 3
(C) s + 3 = 12
(D) 3s = 12

43. A solution that does not change the colour of either red or blue litmus is:

(A) neutral
(B) an indicator
(C) strongly basic
(D) strongly acidic

44. The reciprocal (multiplicative inverse) of 4/9 is:

(A) 4/9
(B) 1/4
(C) 9/4
(D) -4/9

45. During photosynthesis the gas released by green plants into the air is:

(A) water vapour
(B) carbon dioxide
(C) nitrogen
(D) oxygen

46. The value of x in 4x - 6 = 10 is:

(A) x = 4
(B) x = 16
(C) x = 4.75
(D) x = 1

47. The acid present in the human stomach that helps digest food is:

(A) hydrochloric acid
(B) lactic acid
(C) citric acid
(D) sulphuric acid

48. The difference 5/6 - 1/6 equals:

- **(A)** 1
- **(B)** 6/6
- **(C)** 4/6 = 2/3
- **(D)** 4/0

49. Nutrients such as nitrogen, phosphorus and potassium are returned to farm soil by adding:

- **(A)** plain water
- **(B)** fertilisers or manure
- **(C)** more sunlight
- **(D)** extra carbon dioxide

50. If twice a number decreased by 5 equals 9, the number is:

- **(A)** 7
- **(B)** 14
- **(C)** 4.5
- **(D)** 2

51. An antacid tablet relieves acidity in the stomach because it is:

- **(A)** a mild base
- **(B)** a strong acid
- **(C)** an indicator
- **(D)** a neutral salt

52. How many rational numbers lie between 1/4 and 1/2?

- **(A)** exactly three
- **(B)** none
- **(C)** infinitely many
- **(D)** exactly one

53. A plant kept completely in the dark for several days fails to make food because it lacks:

- **(A)** light energy
- **(B)** water
- **(C)** carbon dioxide
- **(D)** chlorophyll

54. The solution of the equation 6 = 2x is:

- **(A)** x = 4
- **(B)** x = 8
- **(C)** x = 3
- **(D)** x = 12

55. Which one of these gives the same colour with both turmeric and litmus as a base does?

- **(A)** vinegar
- **(B)** another basic solution such as washing soda
- **(C)** sugar solution
- **(D)** lemon juice

56. Expressed as a rational number, the decimal 0.5 is:

- **(A)** 1/2
- **(B)** 1/5
- **(C)** 5/1
- **(D)** 5/100

57. Organisms that cannot make their own food and depend on others for it are called:

- **(A)** producers
- **(B)** autotrophs
- **(C)** heterotrophs
- **(D)** chlorophytes

58. If 18 sweets are shared so each of x children gets 3, the equation and answer are:

- **(A)** $3x = 18, x = 9$
- **(B)** $3x = 18, x = 6$
- **(C)** $x + 3 = 18, x = 15$
- **(D)** $18x = 3, x = 6$

59. To exactly neutralise an acid you must add a base until the mixture becomes:

- **(A)** neutral
- **(B)** a gas
- **(C)** a strong acid
- **(D)** more acidic

60. The value of (-3/5) + (3/5) is:

- **(A)** 6/5
- **(B)** 0
- **(C)** 3/5
- **(D)** -6/5

61. If one leaf makes 6 units of glucose in sunlight, the total made by 4 such leaves is:

- **(A)** 12 units
- **(B)** 24 units
- **(C)** 18 units
- **(D)** 10 units

62. A leaf makes x units of food; if 2 leaves together make 16 units, then x equals:

- **(A)** 8 units
- **(B)** 14 units
- **(C)** 4 units
- **(D)** 32 units

63. Common salt (table salt), formed from an acid and a base, has the chemical name:

- **(A)** magnesium sulphate
- **(B)** sodium chloride
- **(C)** sodium bicarbonate
- **(D)** calcium carbonate

64. Which rational number is equivalent to 3/4?

- **(A)** 9/12
- **(B)** 6/12
- **(C)** 4/3
- **(D)** 3/12

65. Water absorbed by the roots is carried up to the leaves through tubes called:

- **(A)** phloem
- **(B)** xylem
- **(C)** stomata
- **(D)** nodules

66. If 3 drops of base neutralise 1 drop of acid and 5x drops of base are used for 5 acid drops, then x is:

- **(A)** x = 5
- **(B)** x = 3
- **(C)** x = 1
- **(D)** x = 15

67. If 3 drops of base neutralise 1 drop of a certain acid, then 12 drops of acid need:

- **(A)** 36 drops of base
- **(B)** 9 drops of base
- **(C)** 4 drops of base
- **(D)** 15 drops of base

68. A water tank is 2/3 full; if 1/6 more is added, the tank is now:

- **(A)** 5/6 full
- **(B)** 1/2 full
- **(C)** full
- **(D)** 3/9 full

69. Lichens, often seen on rocks and tree bark, are an example of a partnership between:

- **(A)** two fungi
- **(B)** two parasites
- **(C)** a fungus and an alga
- **(D)** an animal and a plant

70. The value of m in m/2 + 1 = 5 is:

- **(A)** m = 8
- **(B)** m = 10
- **(C)** m = 4
- **(D)** m = 12

71. Phenolphthalein, a common indicator, stays colourless in acid but in a base turns:

(A) pink
(B) yellow
(C) black
(D) blue

72. The quotient (2/5) / (4/5) in simplest form is:

(A) 2/4
(B) 5/2
(C) 1/2
(D) 8/25

73. A relationship in which two different organisms live together and both benefit is called:

(A) predation
(B) competition
(C) parasitism
(D) symbiosis

74. If 5 is added to twice a number the result is 17. The number is:

(A) 6
(B) 11
(C) 8.5
(D) 12

75. Rain that becomes acidic due to gases from burning fuels is called:

(A) hard water
(B) distilled water
(C) mineral water
(D) acid rain

76. The rational number 1 lies between which pair using halves?

(A) 0 and 1/2
(B) 3/2 and 5/2
(C) 2/2 and 4/2
(D) 1/2 and 3/2

77. The cell organelles inside leaf cells that actually contain chlorophyll are the:

(A) mitochondria
(B) vacuoles
(C) nucleus
(D) chloroplasts

78. Solving the equation 9 - x = 4 gives:

(A) x = 13
(B) x = 5
(C) x = -5
(D) x = 36

79. Why does adding lime to a lake harmed by acid rain help fish survive?

- **(A)** lime makes the water more acidic
- **(B)** lime is a base that neutralises the acid in the water
- **(C)** lime is a food for the fish
- **(D)** lime adds oxygen for the fish

80. If a leaf covers 3/8 of a sheet and another covers 1/8, together they cover:

- **(A)** the whole sheet
- **(B)** 1/2 of the sheet
- **(C)** 4/16
- **(D)** 3/8

81. If a leaf produces 9 bubbles of oxygen each minute in bright light, in 5 minutes it makes:

- **(A)** 54 bubbles
- **(B)** 45 bubbles
- **(C)** 14 bubbles
- **(D)** 40 bubbles

82. The perimeter of a square is 4s. If a square garden's perimeter is 28 m, its side s is:

- **(A)** 14 m
- **(B)** 7 m
- **(C)** 24 m
- **(D)** 112 m

83. Two strips of litmus are dipped in vinegar. Which colour change is correct?

- **(A)** blue litmus turns red, red litmus stays red
- **(B)** red litmus turns blue, blue stays blue
- **(C)** neither strip changes colour
- **(D)** both strips turn green

84. Which list is arranged from smallest to largest: -1/2, 0, 1/2 ?

- **(A)** 1/2, -1/2, 0
- **(B)** 1/2, 0, -1/2
- **(C)** 0, -1/2, 1/2
- **(D)** -1/2, 0, 1/2

85. Why is the lower surface of most leaves usually lighter green than the upper surface?

- **(A)** the lower surface has no veins
- **(B)** the lower surface makes more food
- **(C)** the upper surface has more chloroplasts to catch sunlight
- **(D)** the lower surface stores all the water

86. Half of a number, added to 6, gives 10. The number is:

- **(A)** 20
- **(B)** 32
- **(C)** 4
- **(D)** 8

87. Baking soda used in cooking and as a mild antacid is chemically a:

- **(A)** strong base like caustic soda
- **(B)** salt that gives a basic solution
- **(C)** neutral like table salt
- **(D)** strong acid

88. The product of a rational number and its reciprocal is always:

- **(A)** 0
- **(B)** the number itself
- **(C)** its negative
- **(D)** 1

89. Insectivorous plants are usually found growing in soil that is poor in:

- **(A)** sunlight
- **(B)** carbon
- **(C)** nitrogen
- **(D)** oxygen

90. If x + x + x = 18, then x equals:

- **(A)** 3
- **(B)** 6
- **(C)** 54
- **(D)** 9

91. The bee sting injects an acidic liquid; the soothing agent is therefore one that is:

- **(A)** acidic
- **(B)** oily
- **(C)** neutral
- **(D)** basic

92. Adding 1/3 to a number gives 5/6. The number (as a rational) is:

- **(A)** 7/6
- **(B)** 1/2
- **(C)** 4/3
- **(D)** 1/3

93. The main reason a desert plant such as a cactus has very small, spine-like leaves is to:

- **(A)** store extra starch
- **(B)** reduce water loss
- **(C)** trap more sunlight
- **(D)** attract more insects

94. Check: is x = 4 a solution of 5x - 3 = 17?

- **(A)** No, because x must be 3
- **(B)** Cannot be checked without a graph
- **(C)** Yes, because 5x4 - 3 = 17
- **(D)** No, because 5x4 - 3 = 20

95. A wasp sting is alkaline (basic), so the correct soothing agent is a mild:

- **(A)** base like baking soda
- **(B)** salt solution
- **(C)** acid like vinegar
- **(D)** neutral like water

96. Acid is mixed so it is 1/4 of a bottle and water is 1/2; the empty space is:

- **(A)** 1/2 of the bottle
- **(B)** 1/4 of the bottle
- **(C)** nothing - it is full
- **(D)** 3/4 of the bottle

97. Two leaves together release 14 units of oxygen; if one leaf released 6 units, the other released:

- **(A)** 6 units
- **(B)** 20 units
- **(C)** 8 units
- **(D)** 7 units

98. When 4 is subtracted from three times a number, the result is 11. The number is:

- **(A)** 9
- **(B)** 5
- **(C)** 2.33
- **(D)** 15

99. If pure water leaves litmus unchanged but adding a little lemon turns blue litmus red, lemon is:

- **(A)** more acidic than water
- **(B)** more basic than water
- **(C)** exactly as neutral as water
- **(D)** a salt solution

100. The rational number lying exactly midway between 1/2 and 3/2 is:

- **(A)** 1/2
- **(B)** 2
- **(C)** 3/4
- **(D)** 1